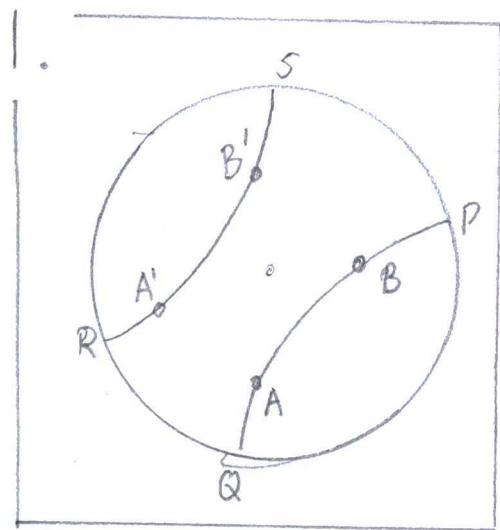


Chapter 7: Independence of the Parallel Postulate:

Projects:



"Congruence Axiom #1"

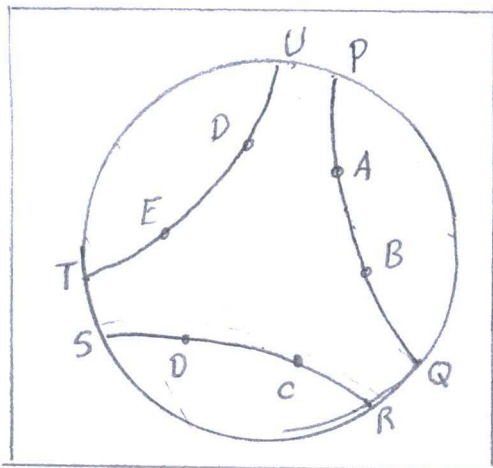
Congruence Axiom #1:

Euclidean:

(pg 119) "Axiom C-1"

"IF A and B are distinct points and if A' is any point, then for each ray r emanating from A' there is a unique point B' on r such that $B' \neq A'$ and $AB \cong A'B'$ "

Hyperbolic: IF A and B are distinct points and if A' is any point, then for each ray r emanating from A', there is a unique point B' on r such that $B' \neq A'$ and $d(AB) \approx d(A'B')$
by Poincare length $d(AB) = |\log(\frac{AP}{BP} \frac{BQ}{AQ})|$
 $= |\log(\frac{AP}{BP} \frac{BQ}{AQ})|$



"Congruence Axiom #2"

Congruence Axiom #2

Euclidean:

(pg 120) "Axiom C-2"

"If $AB \cong CD$ and $AB \cong EF$,
then $CD \cong EF$. Moreover, every
segment is congruent to itself."

Hyperbolic: "If $d(AB) \cong d(CD)$ and
 $d(AB) \cong d(EF)$, then $d(CD) \cong d(EF)$."

Moreover, every segment is
congruent to itself."

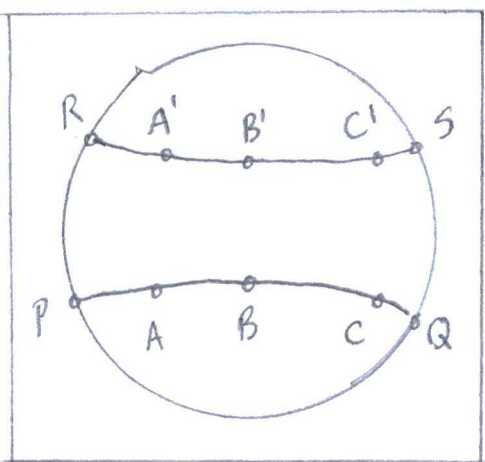
Congruence Axiom #3

Euclidean:

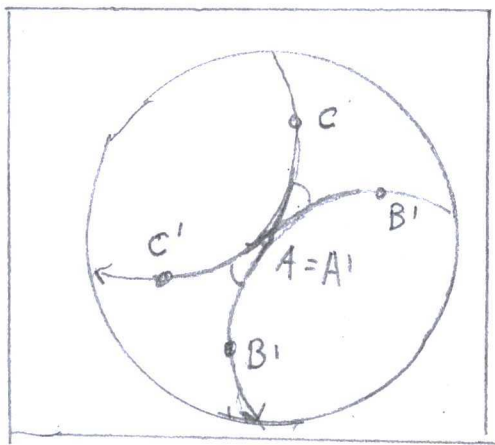
(pg 120) Axiom C-3

"If $A*B*C$, $A'*B'*C'$, $AB \cong A'B'$,
and $BC \cong B'C'$, then $AC \cong A'C'$."

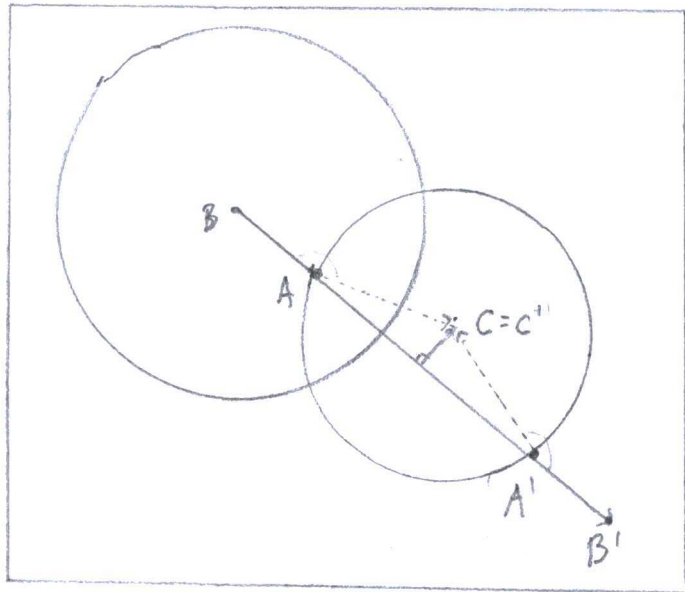
Hyperbolic: "If $A*B*C$, $A'*B'*C'$,
 $d(AB) \cong d(A'B')$ and $d(BC) \cong d(B'C')$,
then $d(AC) \cong d(A'C')$."



"Congruence Axiom #3"



" $\angle BAC$ through center"



" $\angle BAC$ not through center"
"Congruence Axiom #4"

Congruence Axiom #4:

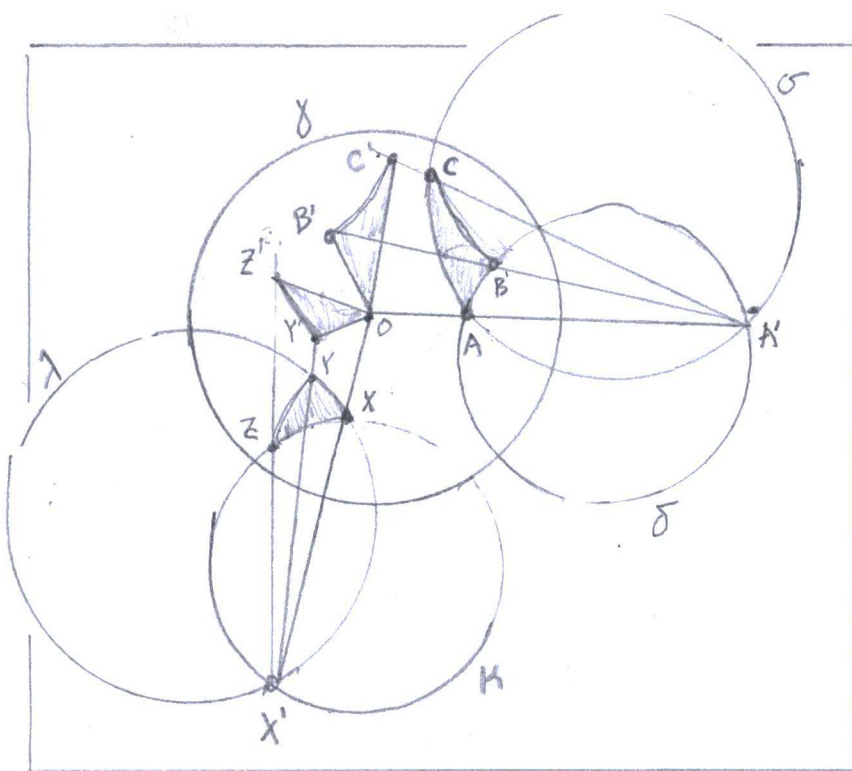
Euclidean:

(pg 120) "Axiom C-4"

"Given any $\angle BAC$ and given any ray $\overrightarrow{A'B'}$ emanating from a point A' , then there is a unique ray $\overrightarrow{A'C'}$ on a given side of line $\overleftrightarrow{A'B'}$ such that $\angle B'A'C' \cong \angle BAC$."

Hyperbolic: "Given any $\angle BAC$

and given any ray $\overrightarrow{A'B'}$ emanating from a point A' then there is a unique ray $\overrightarrow{A'C'}$ on a given side of line $\overleftrightarrow{A'B'}$ such that $\angle B'A'C' \cong \angle BAC$ "



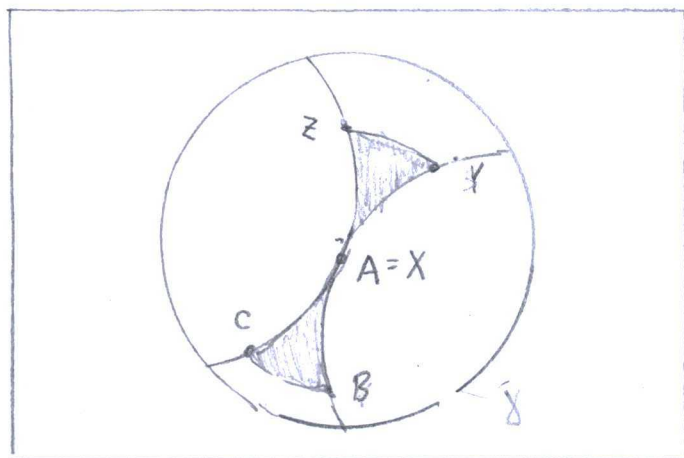
Congruence Axiom #6 :

Euclidean:

(Pg 121) Axiom C-6 (SAS):

"If two sides and the included angle of one triangle are congruent, respectively, to two sides and the included angle of another triangle, then the two triangles are congruent."

" $\triangle ABC$ not through center,
 SAS ($\angle A \cong \angle X$, $d(AC) \cong d(XZ)$, $d(AB) \cong d(XY)$)"



" $\triangle ABC$ through center"
 'Congruence Axiom #6'

Hyperbolic:

"If two sides and the included angle of one triangle are congruent, respectively, to two sides and the included angle, then the two triangles are congruent."

Proof: Hypothesis

Evidence

1) Two Poincare triangles $\triangle ABC$ and $\triangle XYZ$ in $\mathcal{X}$ such $d(AC) \cong d(XZ)$, $\angle A \cong \angle X$, and $d(AB) \cong d(XY)$

RAA conjecture

Case 1 ($A \neq O$):

A) Unique circle σ orthogonal to $\mathcal{X}$

Proposition 7.11

B) A maps to O by inversion A'

(pg 314) "Definition-Inverse"

C) B maps to B' from A

Proposition 7.5

D) Unique circle σ orthogonal to $\mathcal{X}$

Proposition 7.11

E) C maps to C' from point A

Proposition 7.5

F) $\triangle ABC \cong \triangle OB'C$

Proposition 7.11

G) Unique circle $\mathcal{K}$ orthogonal to $\mathcal{X}$

Proposition 7.11

H) X maps to O by inversion X'

I) Y maps to Y' by point X'

J) Unique circle λ orthogonal to δ

K) Z maps to Z' by X'

$$L) \triangle XYZ \cong \triangle OY'Z$$

$$M) \triangle OB'C \cong \triangle OY'Z$$

Case 2: $A=O$

$$A) d(AB) = d(XY)$$

$$B) \angle A \cong \angle X$$

$$C) d(AC) \cong d(XZ)$$

$$2) \triangle ABC \cong \triangle XYZ$$

by
 $S.A.S.$

(pg 314) "Definition-Inverse"

Proposition 7.5

Proposition 7.11

Proposition 7.5

Proposition 7.11

Proposition 7.11

(pg 320) "Poincare length"

Proposition 7.9

Proposition 7.10.

RAA result

2. Cite: Yaglom, I. M.

"A simple Non-Euclidean Geometry
and its Physical Basis"

University of Yaroslavl, (1979)

(pg 218) Table I: Nine-Cayley-Klein
Geometries in the Plane:

Measure of Angles	Measure of length		
	Elliptic	Parabolic	Hyperbolic
Elliptic	Elliptic Geometry	Euclidean Geometry	Hyperbolic Geometry
Parabolic (Euclidean)	co-Euclidean Geometry	Galilean Geometry	co-Minkowskian Geometry
Hyperbolic	co-Hyperbolic Geometry	Minkowskian Geometry	Doubly Hyperbolic Geometry

Measurement		
Elliptic	Parabolic	Hyperbolic

